

PAPER_1808_GROSS_PITAEVSKII_VORTEX_SIMULATION_UQFF_AETHER

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PAPER_1808 — Gross-Pitaevskii Vortex Simulation on UQFF Universal Aether Superfluid

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: F — Quantum-Field / Aether Dynamics
Date: July 2026
Status: CLOSED — closes GP vortex simulation gap from 08May2025 folder audit
Source derivations: 08May2025/36. Superfluid Vortex Dynamics, 37. Vortex Quantization, 38. Gross-Pitaevskii Vortex Simulation (all Jan 2026)
Calculator surface:
calculate_gross_pitaevskii_vortex_simulation_UA_superfluid(dataset)

Purpose

The Gross-Pitaevskii equation (GPE) governs the dynamics of Bose-Einstein condensates and superfluids, including quantized vortex formation. In UQFF, the Universal Aether [UA] is a cosmic-scale quantum superfluid at $\rho_{\text{vac,UA}} \approx 7.09 \times 10^{-30} \text{ J/m}^3$, and vortices in this medium model cosmic magnetic strings, wormhole throats, and gravitational-wave damping structures. This paper wires the GP vortex simulation as a dedicated public calculator surface with the canonical UQFF primitive set.

The Gross-Pitaevskii equation on [UA]

Time-dependent GPE for the [UA] superfluid condensate wavefunction $\psi(r,t)$:

$$i \cdot \hbar \cdot \partial \psi / \partial t = -(\hbar^2 / 2m_{\text{eff}}) \cdot \nabla^2 \psi + V_{\text{ext}} \cdot \psi + g \cdot |\psi|^2 \cdot \psi - \mu \cdot \psi$$

- where:
- $m_{\text{eff}} = \sqrt{(\rho_{\text{vac,UA}} \cdot G / c^2)} \approx \text{Planck-mass-scale (aether quantum effective mass)}$
 - g = repulsive interaction strength on [UA]
 - μ = chemical potential
 - V_{ext} = external potential (gravitational + magnetic + phonon coupling)

Quantized circulation on [UA]

Wavefunction $\psi = \sqrt{\rho} \cdot \exp(i \cdot \theta)$ with velocity $v_s = (\hbar / m_{\text{eff}}) \cdot \nabla \theta$. Around any closed loop enclosing a vortex:

$$\kappa = \oint v_s \cdot dl = (\hbar / m_{\text{eff}}) \cdot 2\pi \cdot n = (h / m_{\text{eff}}) \cdot n \quad (n = \text{integer winding})$$

For [UA] with $m_{\text{eff}} \sim$ Planck-mass scale, single-quantum circulation:

```
``  
κ_UQFF = h / m_eff = h · c / √(ρ_vac, [UA] · G)  
≈ 103 m2/s (cosmic vortex circulation)  
``
```

Canonical primitive corrections

Under the current UQFF 9-primitive set:

F_TRZ negentropic damping:

```
``  
n_eff = n · (1 - F_TRZ) = n · 0.9  
E_v,UQFF = E_v · (1 - F_TRZ) · (1 + β_i · S_26 · Φ_res)  
``
```

SCm superconductive pinning:

```
``  
v_s,UQFF = v_s · (1 - B/B_crit) with B_crit ~ 1011 T for [SCm]  
``
```

Magnetic-string tension via U_m:

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``  
E_v,UQFF += U_m · ln(b/ξ)  
U_m = μ_j / r · (1 - exp(-γ·t·cos(π·t_n)))  
``
```

Vortex energy per unit length (UQFF-corrected)

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``  
E_v = (π · ρ_vac, [UA] · ■2 / m_eff) · ln(b/ξ) · (1 - F_TRZ) · (1 + β_i ·  
S_26_scale · Φ_res)  
``
```

For b (system size) = 10² m (galactic scale), ξ (core size) ~ 10¹ m:

```
``  
E_v ≈ 103 · 101 · 24 · 0.9 · 1.048 ≈ 102 J/m  
``
```

Cosmic-vortex energies at this scale correspond to observed cosmic strings / magnetic-string tensions in galaxy filaments — consistent with cosmic-web observations.

Numerical GP simulation (from 08May2025/38.)

The source derivation implemented a 2D split-step Fourier GP simulation:

- Grid: 128 × 128
- Box: L = 20
- Time step: dt = 0.1
- Total steps: 100
- Interaction: g = 100, μ = 1
- Winding: n = 1 (single quantum)

Density profile shows:

- Vortex core with $\rho \rightarrow 0$ at center
- Surrounding superfluid at $\rho = \mu/g$
- Azimuthal velocity $v(r) = (\hbar \cdot n)/(m_{\text{eff}} \cdot r) \cdot \phi$ (1/r decay)

The simulation numerically confirms the analytical quantization + vortex structure. UQFF extensions add F_TRZ damping and SCm pinning as small perturbations to the classical GP solution.

Cross-references to existing wiring

- **aether_superfluid** and **gross_pitaevskii** paradox dispatches → already wired (PAPER_1055 cMERA entanglement, PAPER_1053 Swampland)
- **vortex quantization** → PAPER_337 (frozen-planet kinetic powering, uses similar quantization)
- **UQFF Superfluid dynamics** → PAPER_1809 (Aether Superfluid Dynamics companion whitepaper)

Verification against 08May2025 source

The three source derivations (files 36, 37, 38 from Jan 2026) use the pre-canonical $(1 \pm f_{\text{TRZ}})$ approach. This paper updates to the full 9-primitive framework while preserving the same GP-based physics (quantization, single-valued phase, vortex-line energy). Numerical GP results (vortex core structure, 1/r velocity, energy density) unchanged.

NOT REPLACEMENT

Standard GP theory (Pitaevskii 1961, Gross 1961) provides the SM analog for BEC vortex dynamics. UQFF extends GP to cosmological [UA] superfluid without replacing the underlying quantization mechanism. Residuals reported honestly per Rule 7.

Reference

- Source derivations: 08May2025/36., 37., 38. (Jan 2026)
- Companion 08May2025 closures: PAPER_1807 (NGC 2014/2020), PAPER_1809 (Aether Superfluid Dynamics)
- Related repository papers: PAPER_1053 (Swampland conjectures), PAPER_1055 (cMERA entanglement RG), PAPER_1015 (SCm DM halos NFW)

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